AI 507: Artificial Intelligence and Society

#7 Human-Chatbot interaction

Associate Prof. Dr. Lena Frischlich

Flashlight

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Flashlight

Algorithmic recommendation

Basic mechanisms

Collaborative filtering

Content-based filtering

Hybrid approaches

Deep-learning

Audience perspective

Attitudes towards algorithmic recommendations

Engagement & algorithmic recommendations

User-control & for the social good

How to embed public values

Algorithmic literacy

Facets

Distribution

Over-trusting

Micro-targeting

Basic idea

Motivated cognition

Cambridge Analytica

Use in political campaigns

Selective exposure

Filterbubbles

Echo-chambers

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Any questions on that?

Then let’s get started!

After today’s lecture, you will hopefully…

- ❑ …have a better understanding of AI-generated communication
- ❑ …have explored how AI-chatbots can endanger democratic discourse
- ❑ …have learned more about AI-chatbots for democracy
- ❑ …have discussed intimate relations with AI-chatbots

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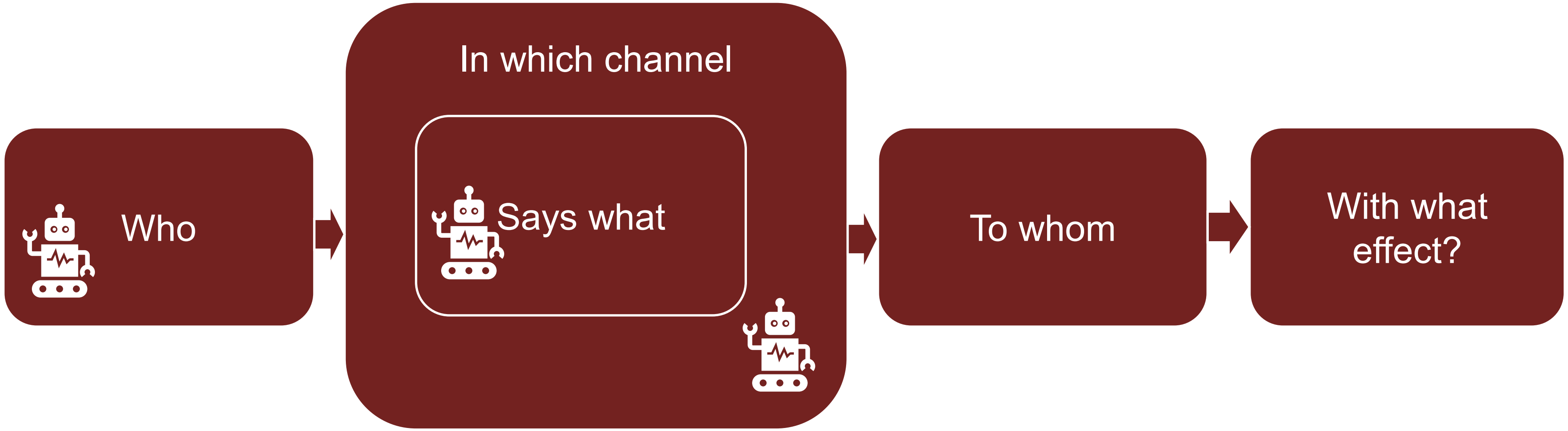
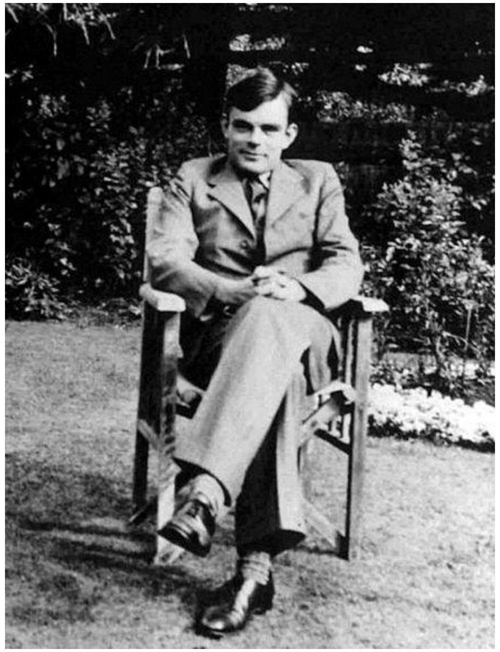
Algorithmic communication

Rember Lasswell's formula of communication? (Lasswell; 1948; Turing, 1950; Reeves & Nass, 1996; Smith, 2018; You et al., 2022)




Alan Turing, 1938 via Wikipedia

- Referred to as chatbots or "social bots"
- Automatisation of social processes, specifically communication
- Builds on automatisation as well as algorithmisation → processes and phenomena such as the media equation/ CASA or biases in algorithmic interaction apply here too!
- Evaluation of their performance rooted in the idea of the Turing test

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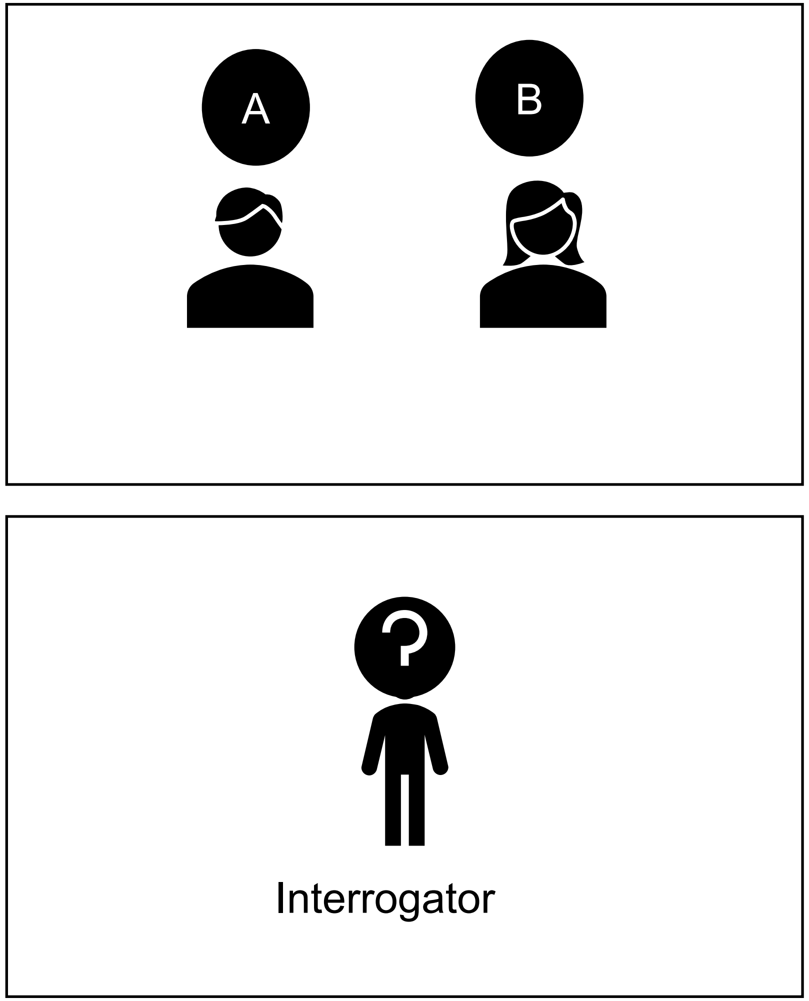
The Turing Test (Turing, 1950)



- The Interrogator has the task to find out who the woman is
  - To this end, he can ask questions – but only as text
  - The questions will be answered – only as text
- The woman wants the interrogator to identify her as a woman in the end correctly
  (so honesty is a good strategy)
- The man wants to deceive the interrogator
  (so it may make sense to lie)
- Artificial "intelligence" is, when the interrogator is mistaken just as often when a bot instead of a man tries to deceive the interrogator
- Lying at the heard of the optimization criteria?

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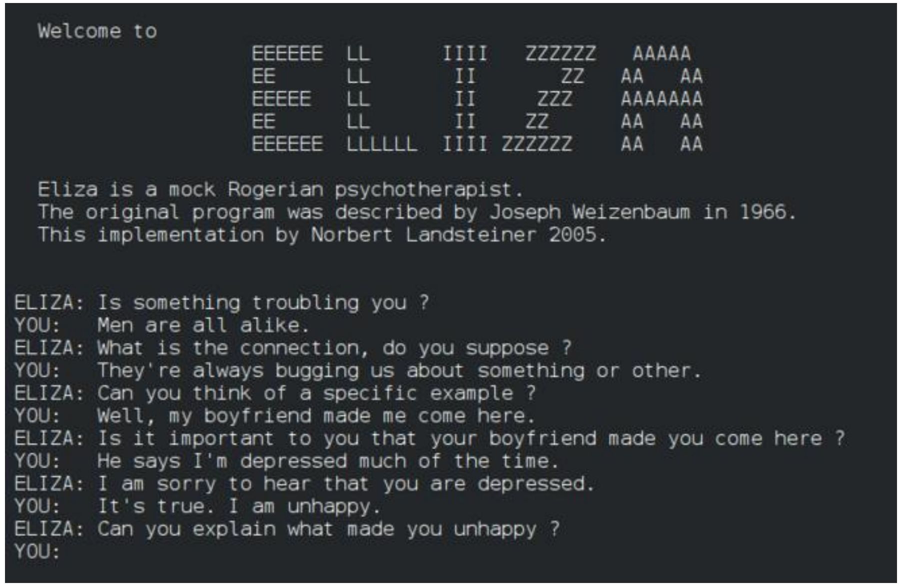
ELIZA (Weizenbaum, 1966)

Joseph Weizenbaum, 2005 via Wikipedia

- The first “famous” chatbot
- Computer program (or better programs) to communicate with the user
- Scanned the written input for keywords and responded according to fixed scripts developed based on Carl Rogers' talk therapy (→ symbolic AI, fixed rules)
- Still, some needed a while to figure out that “the doctor” was a computer program
- You can try it yourself at http://www.med-ai.com/models/eliza.html.de



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In the last decade: Increasing interest in social bots (Frischlich et al., 2020; Grimme et al., 2017)

- Social-bots = super-ordinate concept, which summarizes different types of (semi-) automatic agents. These agents are designed to fulfill a specific purpose by means of one- or many-sided communication in online media, particularly social media
- Requires a communicative infrastructure

User
- Individual
- Group
- Organisation
- State

Steering strategy

Orchestration
- fully automated
- partially automated
- manually

Acts as

Account
- Authentic
- Compromised
- fake

Visible

Actions
- Amplification
- Connection
- Creation

Heavily dependent on access and availability

Initially: Not very "intelligent" (Assenmacher et al., 2020)

- Emerged from a project around the German parliamentary election 2017
- Market analysis of Clearnet and Darknet markets: What kind of social bots can you buy (*n* = 1102)
  - Most bots were traded on Clearnet markets (~ 9:1)
  - Mostly for large-scale platforms, particularly those with closed APIs
  - Mostly amplification (like/ share inflation, fake followers)

Analysis of openly available bot code 2009-2019 (*N* = 45,018 repositories)

  - Mostly simple functions
  - Few artificial intelligence beyond Markov chains
  - Mostly for large-scale platforms, particularly those with open APIs
  - But then came transformer models and the GPT-revolution

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And now: GPT is…

(Hulum & Vestergaard, 2024; www.semrush.com; https://backlinko.com; https://originality.ai/)

9th most

popular website worldwide

5.19 billion

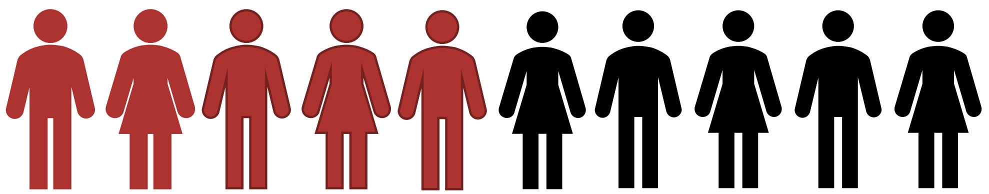
visitors in February 2025

62 million

Used Deep Seek

36 million

Used Hugging Face /2024



1 in 2 Danish workers has already used ChatGPT
Younger-higher educated men are most likely to report usage

Barriers to adoption
(→ technology acceptance model; #5 innovation)
perceived norms &
perceived need for training

What are your questions?

Risks of AI-dissemination Social bots and the manipulation of the democratic discourse

Increasing concerns about social bots

(Assenmacher et al., 2020; Howard & Kollanyi, 2016; Ferrara et al., 2016)

review articles

DOI:10.1145/2818717

Today's social bots are sophisticated and sometimes menacing. Indeed, their presence can endanger online ecosystems as well as our society.

BY EMILIO FERRARA, ONUR VAROL, CLAYTON DAVIS, FILIPPO MENCZER, AND ALESSANDRO FLAMMINI

The Rise of Social Bots

- In the aftermath of the Brexit vote and Donald Trumps' first election
- Several far-reaching concerns about social bots endangering online discourses, spreading rumours and hoaxes, affecting stock markets, engaging in astroturfing campaigns and stealing private data
- Partially due to peoples' lack of scrutiny: 20% accept friends requests from everyone
- Led to wild attempts in "detecting" bots – e.g., using high tweeting activity as an indicator

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Coordinated inauthentic behaviour (Giglietto et al., 2020)

- Examination of the coordinated sharing of news on Facebook in the 2018' Italian general election and the 2019 EU-election (*~230k posts*)
- Untrustworthy entities share content faster than trustworthy entities
- 20-times more likely to be in the "coordinated" networks than in the uncoordinated ones
- Typically: very few accounts are responsible for most of the deviant content

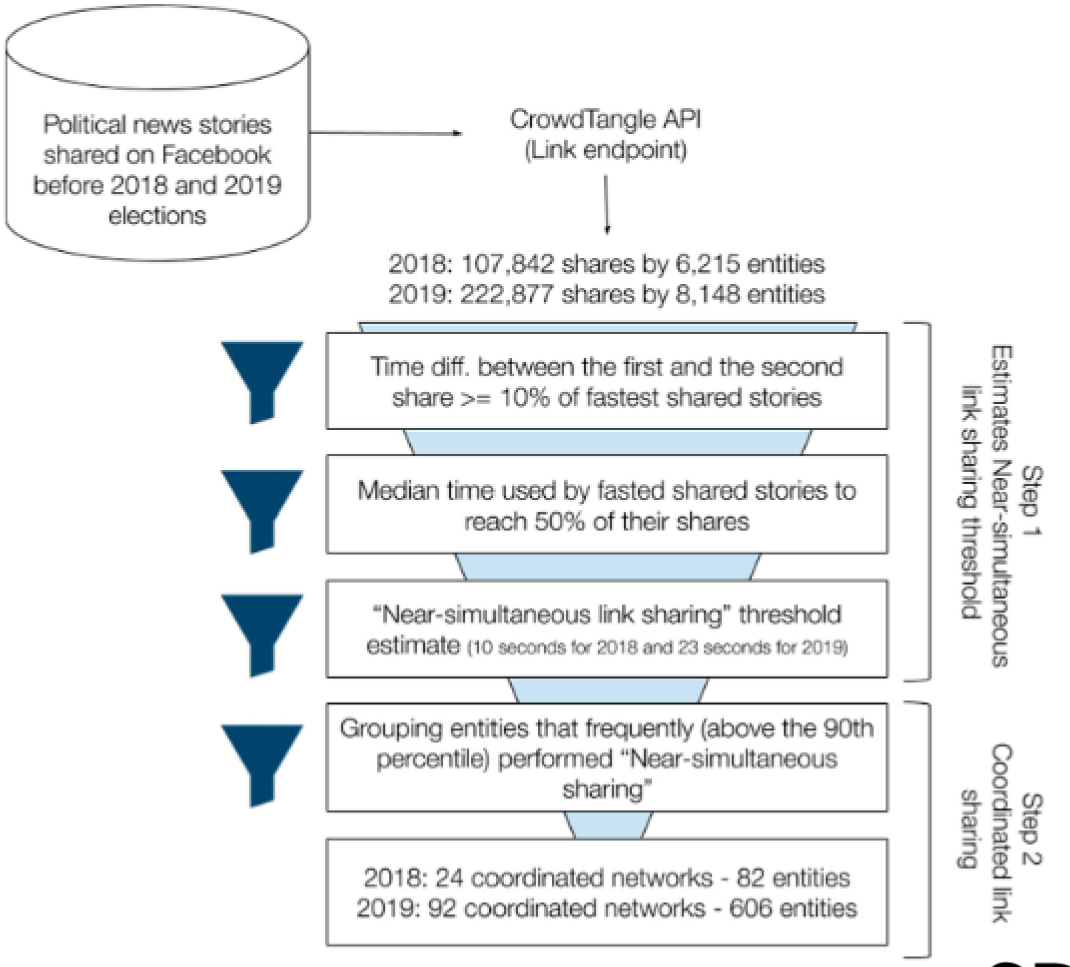


Figure 1. Funnel of coordinated link sharing.

Some numbers about superspreaders

3% of accounts tweeting about the US-general election 2024

10% of accounts sharing images on Twitter about the US-general election 2024

80% of AI-generated content

- More likely to be right-wing leaning
- More likely to be premium subscribers
- More likely to exhibit inauthentic patterns

0,2% of users in Danish public Facebook groups

- 68% use a male name
- Assumption that users > 40 account for 80%

0,3 % of accounts tweeting about the US-general election 2020

80% of misinformation

- Slightly more likely to be women
- On average ~60 years
- Republicans
- White

Why bot or not is not the question (Grimme et al., 2020)

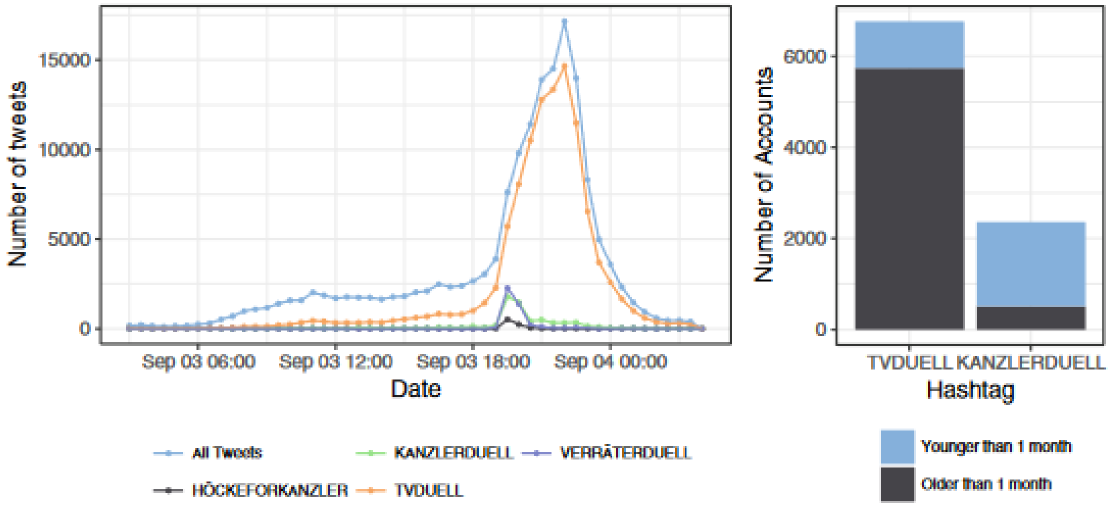


Fig. 8. Important indicators for the first case-study. The figure on the left hand side shows a time series of the activities during the TV debate. The figure on the right hand side shows the proportion of new and old accounts active for two hashtags.

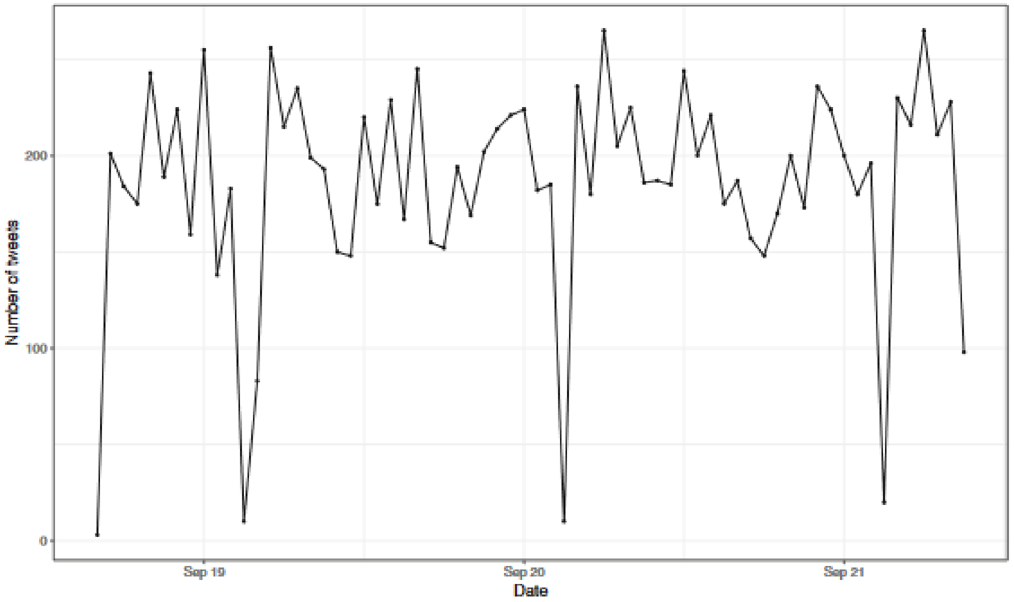


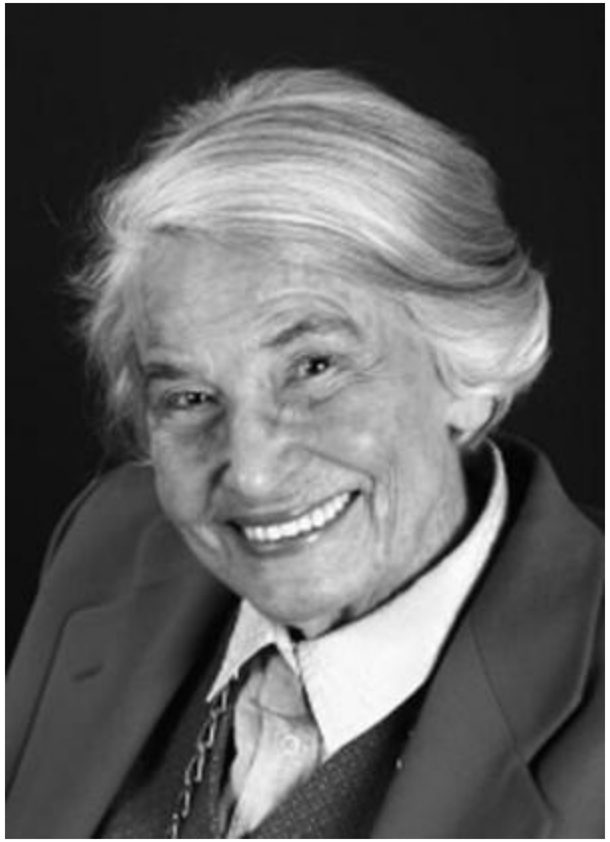
Fig. 10. Number of tweets for #freiewaehler over time.

What die Grimme et al.
find out?

But does it matter? (Noelle-Neumann, 1974)

The spiral of silence theory

- Basic idea: People are social animals
- Fear of isolation → quasi-statistical organ to detect majority opinions around them
- Those who perceive themselves to be in the majority are more likely to speak up, those who are in the minority are less likely to do so
- E.g., disliking Deadpool



Noelle-Neumann (privat) via http://blexkom.halemverlag.de/elisabeth-noelle-neumann/

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Empirical evidence (Hayes & Matthes, 2014; Neubaum & Krämer, 2017; Ross et al., 2019)

- Yes: a weak link between the perceived congruence of public opinion and the willingness to speak up
- But: Highly convinced people who feel they are morally correct are generally more willing to express their opinion
- People also might search for more information instead of remaining silent (→ Dissonance reduction #7 recommender)
- In short, individual differences and a fragmented media system weaken the spiral of silence
- But: Social media discourse is (mis-)interpreted as “public opinion” and can lead to spirals of silence
- Agent-based models show that in high-polarized settings, a consistent minority of 2-4% is already enough to tip over the opinion climate in two out of three discussions

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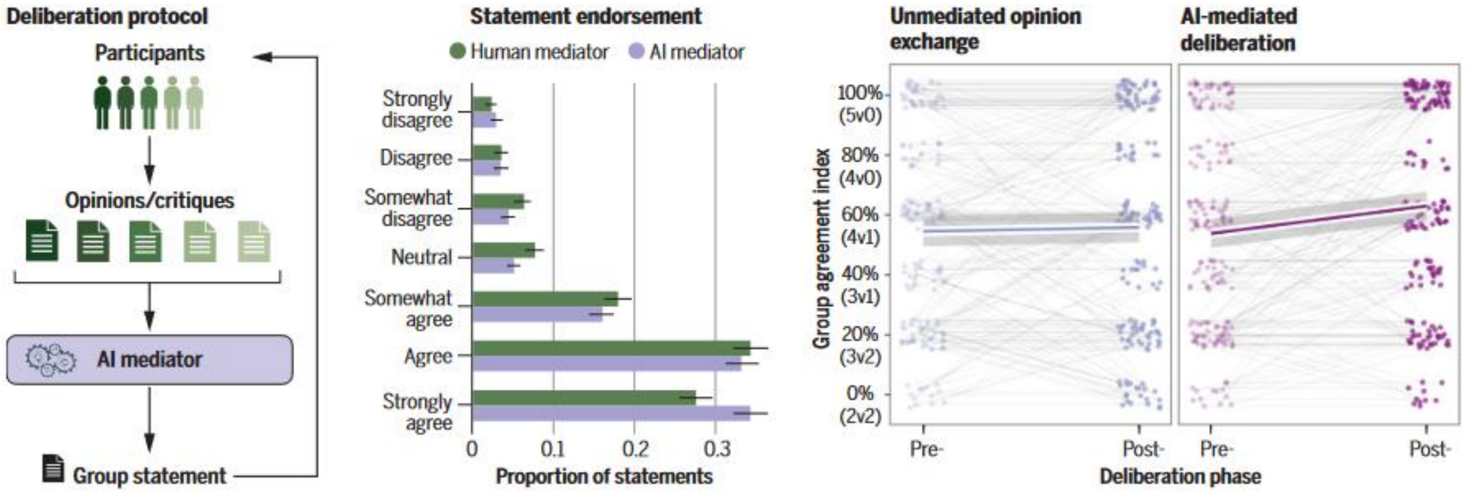
What are your questions?

Social bots to support democracy

AI supporting democracy(1) (Tessler et al., 2024)

- Supporting constructive debates
  - LLM-based system generated group statements based on personal opinions and critiques from individual users
  - Aim: maximizing group approval ratings
  - Group members consistently preferred group opinion statements generated by the *Habermas Machine* over those written by human mediators



AI helps people find common ground in collective deliberation. (Left) The AI mediator uses participants' opinions to generate group statements and iteratively refines those statements through participants' critiques. (Middle) Statements from the AI mediator (purple) garner stronger endorsement than those written by a human mediator (orange). (Right) AI mediation leaves groups less divided after deliberation, whereas simply sharing opinions with others does not.

Tessler *et al.*, *Science* **386**, 289 (2024) 18 October 2024 **1 of 1**

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AI supporting democracy (2) (Tessler et al., 2024)

- AI-generated counters-speech
  - Addressing n = 454 users that posted at least 30 hateful comments on Reddit (/rMensRights and r/TooAfraidTooAsk)
  - Bot replied to hateful comments through reminding the user of civic communication norms or calling for empathy with the target

BILEWICZ ET AL. AGGRESSIVE BEHAVIOR WILEY 263

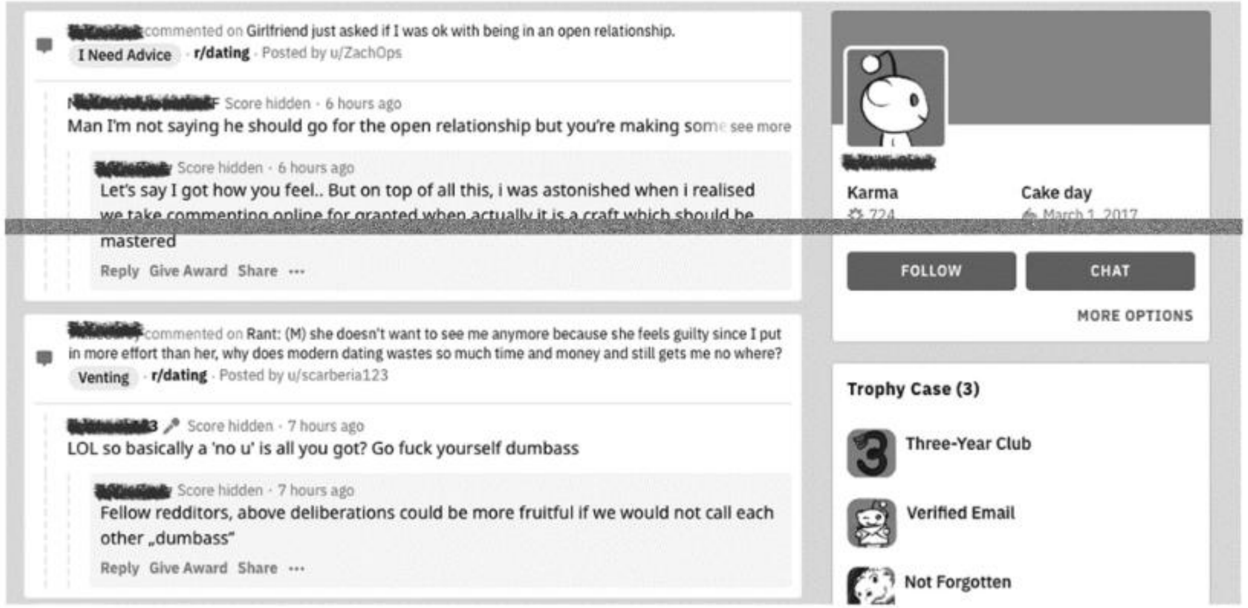

FIGURE 1 The profile of the artificial bot used in the interventions (with sample comments/messages)

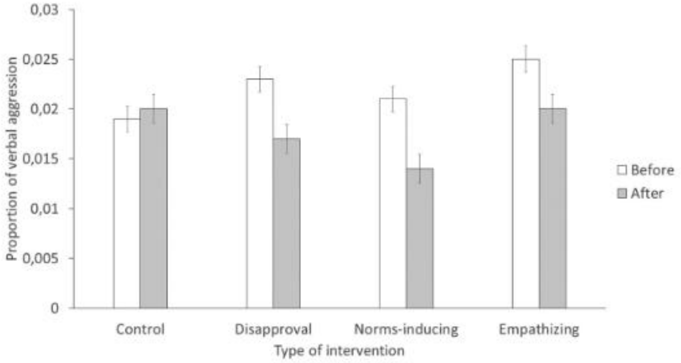

FIGURE 3 Proportion of verbal aggression to all comments before and after disapproval, norms-inducing and empathizing interventions, as well as in the control condition (without intervention)

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AI supporting democracy (3) (Lokot & Diakopolous, 2016)

- In media institutions, bots are used for various purposes
- Already in 2015: Twitter bots for the distribution of own messages
- But also: monitoring other websites, archives, hashtags, etc.
- Bots cover specific niche topics
- They curate and aggregate content and link different platforms (e.g. Reddit & Twitter)
- Their main purpose is to inform

Bots

We're exploring how bots can help the BBC reach new audiences and simplify workflow for our editors and journalists.

Contents

Published: 25 August 2020

Aims

How can we leverage bot technologies to reach new audiences on messaging platforms and social media?

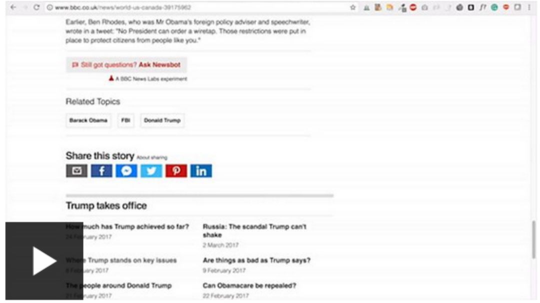
So far we have built bots for Facebook, Twitter, and Telegram in collaboration with the BBC's Visual Journalism and World Service teams. These bots have variously allowed us to automate our coverage of the EU Referendum, reach audiences where World Service websites are blocked and engage readers who wouldn't normally visit our sites.

We're currently exploring new opportunities offered by artificial intelligence, machine learning and speech recognition technologies.

Individual bots

In-Article Chatbot

We piloted the BBC's first **in-article chatbot** by adding a conversational interface to the bottom of BBC News stories. This gave readers the ability to find out more about a topic in a more conversational style. For the trial, we added the explainer bots to articles where the subject domain can be more complex. The conversations were based on pre-scripted material written in a multiple choice format. After a successful pilot, we **adapted the in-article chatbots** for use in the UK's general election.



A demo of our in-article chatbot.

Time for a break!

Intimate human-chatbot relations

Dr. Jessica Szczuka,
University of Duisburg-Essen

Then I hope you now have…

- ❑ …understood how AI-generated communication is transforming journalism
- ❑ …have a better understanding of when we trust AI – and when not
- ❑ …have explored how AI can endanger democratic discourse
- ❑ …have learned more about AI for democracy

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#9 Human-object interactions: Smart things and the web 4.0

Associate Prof. Dr. Lena Frischlich

References

- Assenmacher, D., Clever, L., Frischlich, L., Quandt, T., Trautmann, H., & Grimme, C. (2020). Demystifying social bots: On the intelligence of automated social media actors. *Social Media + Society*, *6*(3), 205630512093926. https://doi.org/10/gg9fts
- Assenmacher, D., Clever, L., Frischlich, L., Quandt, T., Trautmann, H., & Grimme, C. (2020). Demystifying social bots: On the intelligence of automated social media actors. *Social Media + Society*, *6*(3), 205630512093926. https://doi.org/10/gg9fts
- Baribi-Bartov, S., Swire-Thompson, B., & Grinberg, N. (2024). *Supersharers of fake news on Twitter*.
- Bilewicz, M., Tempska, P., Leliwa, G., Dowgiałło, M., Tańska, M., Urbaniak, R., & Wroczyński, M. (2021). Artificial intelligence against hate: Intervention reducing verbal aggression in the social network environment. *Aggressive Behavior*, *47*(3), 260–266. https://doi.org/10.1002/ab.21948
- Chen, Z., Ye, J., Ferrara, E., & Luceri, L. (2025). *Prevalence, sharing patterns, and spreaders of multimodal AI-generated content on X during the 2024 u.s. Presidential election* (arXiv:2502.11248). arXiv. https://doi.org/10.48550/arXiv.2502.11248
- de-Lima-Santos, M.-F. (2022). ProPublica's data journalism: How multidisciplinary teams and hybrid profiles create impactful data stories. *Media and Communication*, *10*(1), 5–15. https://doi.org/10.17645/mac.v10i1.4433
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding* (No. arXiv:1810.04805). arXiv. https://doi.org/10.48550/arXiv.1810.04805
- Dierickx, L., van Dalen, A., Opdahl, A. L., & Linden, C.-G. (2024). Striking the balance in using LLMs for fact-checking: A narrative literature review. *MISDOOM 2024*, 1–15. https://link.springer.com/chapter/10.1007/978-3-031-71210-4_1
- Ferrara, E., Varol, O., Davis, C. A., Menczer, F., & Flammini, A. (2016). The rise of social bots. *Communications of the ACM*, *59*(7), 96–104. https://doi.org/10/f8wdbj
- Fridman, M., Krøvel ,R., & and Palumbo, F. (2023). How (not to) run an ai project in investigative journalism. *Journalism Practice*, *0*(0), 1–18. https://doi.org/10.1080/17512786.2023.2253797
- Frischlich, L., Mede, N. G., & Quandt, T. (2020). The markets of manipulation: The trading of social bots on clearnet and darknet markets. In C. Grimme, M. Preuss, F. W. Takes, & A. Waldherr (Eds.), *Disinformation in open online media* (pp. 89–100). Springer International Publishing. https://doi.org/10/ggpwn3
- Giglietto, F., Righetti, N., Rossi, L., & Marino, G. (2020). It takes a village to manipulate the media: Coordinated link sharing behavior during 2018 and 2019 Italian elections. *Information, Communication & Society*, *23*(6), 867–891. https://doi.org/10/ggsmrg

References

- Grimme, C., Assenmacher, D., & Adam, L. (2018). Changing perspectives: Is it sufficient to detect social bots? In G. Meiselwitz (Ed.), *Social Computing and Social Media. User Experience and Behavior* (Vol. 10913, pp. 445–461). Springer International Publishing. https://doi.org/10.1007/978-3-319-91521-0_32
- Grimme, C., Preuss, M., Adam, L., & Trautmann, H. (2017). Social bots: Human-like by means of human control. *Big Data*, 279–293. https://doi.org/10.1089/big.2017.0044
- Hayes, A. F., & Matthes, J. (2014). *Self-censorship, the spiral of silence, and contemporary political communication* (K. Kenski & K. H. Jamieson, Hrsg.; Bd. 1). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199793471.013.31
- Howard, P. N., & Kollanyi, B. (2016). Bots, #StrongerIn, and #Brexit: Computational propaganda during the UK-EU referendum. *SSRN Electronic Journal*. https://doi.org/10/gf3g8w
- https://backlinko.com/deepseek-stats
- https://originality.ai/blog/huggingface-statistics
- https://www.bbc.co.uk/rdnewslabs/projects/bots/
- https://www.ibm.com/think/topics/transformer-model
- https://www.semrush.com/website/chatgpt.com/overview/
- Humlum, A., & Vestergaard, E. (2024). *The Adoption of ChatGPT* [Discussion paper series]. IZA Institute of Labor Economics.
- Lasswell, H. D. (1948). The structure and function of communication in society. In L. Bryson (Ed.), *The communication of ideas* (pp. 215–228). The Institute for Religious and Social Studies.
- Lokot, T., & Diakopoulos, N. (2016). News bots: Automating news and information dissemination on Twitter. *Digital Journalism*, *4*(6), 682–699. https://doi.org/10.1080/21670811.2015.1081822
- Neubaum, G., & Krämer, N. C. (2017). Monitoring the opinion of the crowd: Psychological mechanisms underlying public opinion perceptions on social media. *Media Psychology*, *20*(3), 502–531. https://doi.org/10/gfbxwf
- Noelle-Neumann, E. (1974). The spiral of silence. *Journal of Communication*, *24*(2), 43–51. https://doi.org/10/b4n6tv
- Ørtoft, A., Neesgaard, A., Rosen Sørensen, M., Thomsen, M., Mørch, T., & Jepsen, Y. (2025). *Angreb & had i den offentlige debat på Facebook*. TryFonden.

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References

- Reeves, B., & Nass, C. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. Cambridge University.
- Ross, B., Pilz, L., Cabrera, B., Brachten, F., Neubaum, G., & Stieglitz, S. (2019). Are social bots a real threat? An agent-based model of the spiral of silence to analyse the impact of manipulative actors in social networks. *European Journal of Information Systems*, *00*(00), 1–19. https://doi.org/10/gf3gmv
- Smith, A. (2018, November 16). Attitudes toward algorithmic decision-making. *Pew Research Center: Internet, Science & Tech*. https://www.pewresearch.org/internet/2018/11/16/attitudes-toward-algorithmic-decision-making/
- Stray, J. (2019). Making artificial intelligence work for investigative journalism. *Digital Journalism*, *7*(8), 1076–1097. https://doi.org/10.1080/21670811.2019.1630289
- Tessler, M. H., Bakker, M. A., Jarrett, D., Sheahan, H., Chadwick, M. J., Koster, R., Evans, G., Campbell-Gillingham, L., Collins, T., Parkes, D. C., Botvinick, M., & Summerfield, C. (2024). AI can help humans find common ground in democratic deliberation. *Science*, *386*(6719), eadq2852. https://doi.org/10.1126/science.adq2852
- Thomson, T. J., Thomas, R., Riedlinger, M., & Matich, P. (2025). *Generative AI and Journalism: Content, Journalistic Perceptions, and Audience Experiences* (p. 13751357 Bytes). figshare. https://doi.org/10.6084/M9.FIGSHARE.28068008
- Thurman, N., Lewis, S. C., & Kunert, J. (2019). Algorithms, automation, and news. *Digital Journalism*, *7*(8), 980–992. https://doi.org/10.1080/21670811.2019.1685395
- Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, *59*(236), 433–460. https://doi.org/10/b262dj
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention is all you need*. 1–11.
- Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. *Communications of the ACM*, *9*(1), 36–45.
- You, S., Yang, C. L., & Li, X. (2022). Algorithmic versus Human Advice: Does Presenting Prediction Performance Matter for Algorithm Appreciation? *Journal of Management Information Systems*, *39*(2), 336–365. https://doi.org/10.1080/07421222.2022.2063553
- Young, M. L., Hermida, A., & Fulda, J. (2018). What makes for great data journalism?: A content analysis of data journalism awards finalists 2012–2015. *Journalism Practice*, *12*(1), 115–135. https://doi.org/10.1080/17512786.2016.1270171